

PROJECT: Scalable Web Application with Load Balancing

Services: EC2, ELB, Auto Scaling, RDS, S3, Route 53

Goal: Deploy a web app that scales based on load, with database and static file hosting.

Skills: VPC, security groups, EC2 launch templates, user data scripts.

Definition: A Scalable Web Application with Load Balancing is a system design where a web application can handle increasing user traffic automatically by adding or removing servers, while a load balancer distributes incoming requests evenly across those servers to ensure high performance and availability.

Scope of the project: Design and deploy a scalable web application using cloud services like EC2, Load Balancer, Auto Scaling, RDS, and S3.

Implement automatic scaling to handle varying user traffic efficiently without manual intervention.

Ensure high availability, fault tolerance, and optimized performance through load balancing and multi-AZ deployment.

Methodologies:

Method Used in the Project: The project uses the Auto Scaling method with Target Tracking Scaling Policy based on Application Load Balancer (ALB) Request Count per Target.

- Implemented load balancing using AWS Elastic Load Balancer to distribute incoming traffic across multiple EC2 instances.
- Used Auto Scaling Groups to dynamically add or remove EC2 instances based on CPU utilization and demand.
- Configured security groups, VPC, and subnets to ensure secure and efficient network communication.
- Integrated RDS for database management and S3 for storing and serving static files.

Services Used in this Project:

1. Amazon EC2 (Elastic Compute Cloud)

- **Definition:** Amazon EC2 (Elastic Compute Cloud) is a service that provides scalable virtual servers in the cloud to run applications.
- **Purpose:** Amazon EC2 (Elastic Compute Cloud) is a service that provides scalable virtual servers in the cloud to run applications.
- **Role in this Project:** EC2 instances handle user requests from the Load Balancer and scale automatically using Auto Scaling when traffic increases.

2.ELB (Elastic Load Balancer)

- **Definition:** Elastic Load Balancer is an AWS service that automatically distributes incoming application traffic across multiple EC2 instances.
- **Purpose:** The ELB is used to distribute user requests evenly to the EC2 instances running the web application.
- **Role in this Project:** The ELB receives traffic from users and forwards it to healthy EC2 instances in the Auto Scaling Group to ensure high availability and performance.

3. Auto Scaling :

- **Definition:** Auto Scaling is an AWS service that automatically increases or decreases the number of EC2 instances based on application demand.
- **Purpose:** Auto Scaling is used to automatically adjust the number of EC2 instances in response to changes in user traffic.
- **Role in this Project:** Auto Scaling ensures the web application maintains performance and availability by launching new instances during high load and terminating them when load decreases.

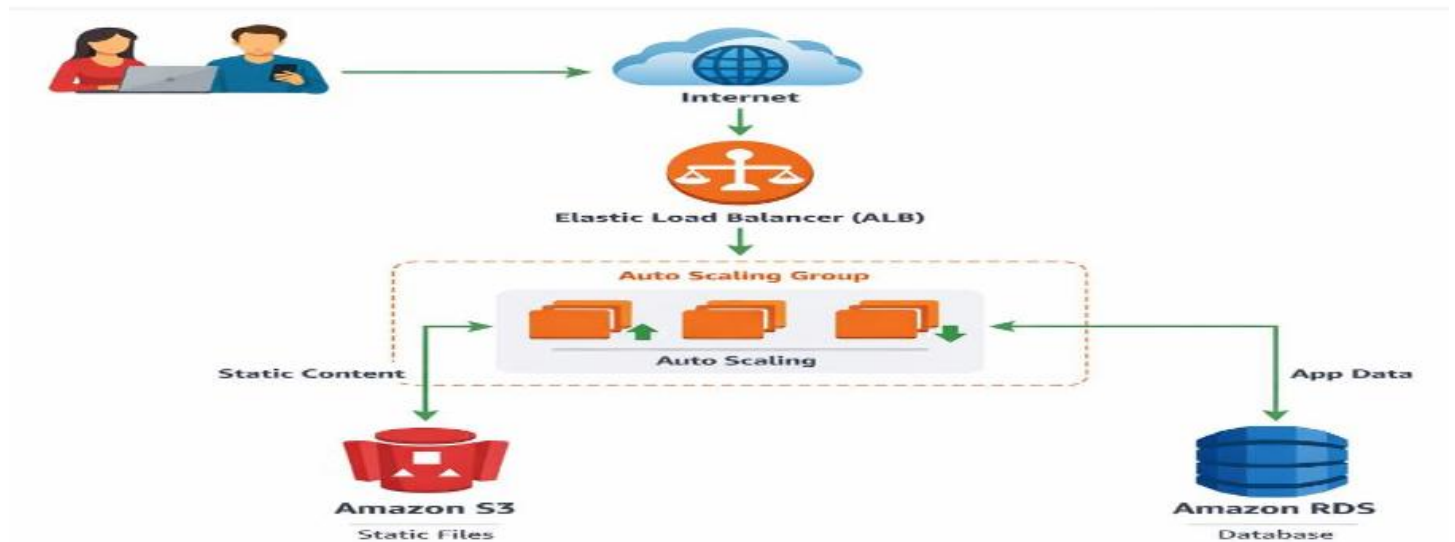
4. RDS (Relational Database Service)

- **Definition:** Amazon RDS is a managed AWS service used to create, operate, and manage relational databases in the cloud.
- **Purpose:** RDS is used to store and manage the web application's database securely and reliably.
- **Role in this Project:** RDS is used to store and manage the web application's database securely and reliably.

5. S3 (Simple Storage Service)

- **Definition:** Amazon S3 is a scalable cloud storage service used to store and retrieve files such as images, videos, and documents.
- **Purpose:** S3 is used to store the static files of the web application like images, CSS, and JavaScript.
- **Role in this Project:** S3 is used to store the static files of the web application like images, CSS, and JavaScript.

Architecture Diagram:



Implementation and Execution of Scalable Web Application with Load Balancing:

Step 1: Created VPC

VPC dashboard < AWS Global View ⓘ Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet

Your VPCs (1/2) Info Last updated less than a minute ago Actions Create VPC

Find VPCs by attribute or tag

Name	VPC ID	State	Encryption c...	Encryption control ...
-	vpc-0f2edfff0b6bd7f6d	Available	-	-
myvpc	vpc-0042a04f0c653ff45	Available	-	-

vpc-0042a04f0c653ff45 / myvpc

STEP 2: Create Public Subnet

VPC dashboard < AWS Global View ⓘ Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets

Subnets (1/1) Info Last updated 2 minutes ago Actions Create subnet

Find subnets by attribute or tag

subnet-0351cc8533bef08dc X Clear filters

Name	Subnet ID	State	VPC	Block Public.
my-Public-Subnet-1	subnet-0351cc8533bef08dc	Available	vpc-0042a04f0c653ff45 myvpc	Off

STEP 3: Create Private Subnet

VPC dashboard < AWS Global View ⓘ Filter by VPC

Virtual private cloud

- Your VPCs
- Subnets

Subnets (1/1) Info Last updated 3 minutes ago Actions Create subnet

Find subnets by attribute or tag

subnet-08a8d36f960b75043 X Clear filters

Name	Subnet ID	State	VPC	Block Public.
my-Private-Subnet-1	subnet-08a8d36f960b75043	Available	vpc-0042a04f0c653ff45 myvpc	Off

STEP 4: Create Public Route Table

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables (1/1)

Find route tables by attribute or tag

rtb-0801e614b48849b26

Clear filters

Last updated 4 minutes ago

Actions

Create route table

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	Public-RT	rtb-0801e614b48849b26	subnet-0351cc8533bef0...	-	No	vpc-0

STEP 5: Create Private Route Table

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Route tables (1/1)

Find route tables by attribute or tag

rtb-00f9150eaff9f2399

Clear filters

Last updated 6 minutes ago

Actions

Create route table

☒	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒	Private-RT	rtb-00f9150eaff9f2399	-	-	No	vpc-0

STEP 6: Created Internet Gateway and Attached to VPC.

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Internet gateways (1/2)

Find internet gateways by attribute or tag

igw-028370a79ad9c6064

Clear filters

Last updated 4 minutes ago

Actions

Create internet gateway

☐	Name	Internet gateway ID	State	VPC ID
☐	-	igw-028370a79ad9c6064	Attached	vpc-0f2edfff0b6bd7f6d
☒	MyApp-IGW	igw-0fb82d5649ac5e547	Attached	vpc-0042a04f0c653ff45 myvpc

igw-0fb82d5649ac5e547 / MyApp-IGW

Details

Tags

Details

Internet gateway ID
igw-0fb82d5649ac5e547

State
Attached

VPC ID
vpc-0042a04f0c653ff45 | myvpc

Owner
074139044318

STEP 7: Create NAT Gateway

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

NAT gateways (1/1)

Find NAT gateways by attribute or tag

nat-1ca58f58709ff3046

Clear filters

Last updated 4 minutes ago

Actions

Create NAT gateway

☒	Name	NAT gateway ID	Connectivity...	State	State message	Availability ...
☒	my-nat-gateway	nat-1ca58f58709ff3046	Public	Available	-	Regional

STEP 8: Create application security group

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Load Balancing

Security Groups (7)

Find security groups by attribute or tag

sg-085deda2d3e6b82f0

Clear filters

Last updated 4 minutes ago

Actions

Export security groups to CSV

Create security group

☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-085deda2d3e6b82f0	ELB-SG	vpc-0042a04f0c653ff45
☐	-	sg-0c9c75d84e804a9ab	WebServer-SG	vpc-0042a04f0c653ff45
☐	-	sg-03dafd13cde760c94	Database-SG	vpc-0042a04f0c653ff45

Inbound Rule:

- ▼ Load Balancing
 - Load Balancers
 - Target Groups
 - Trust Stores
- ▼ Auto Scaling
 - Auto Scaling Groups
- Settings

sg-0cbc75d84e804a9ab - WebServer-SG

Inbound rules (1/1)

sg-09feefa3367a1b49a

Clear filters

< 1 >

⚙

☒	Name	Security group rule ID	IP version	Type	Protocol
☒	-	sg-09feefa3367a1b49a	IPv4	HTTP	TCP

Outbound Rule:

- ▼ Load Balancing
 - Load Balancers
 - Target Groups
 - Trust Stores
- ▼ Auto Scaling
 - Auto Scaling Groups
- Settings

sg-0cbc75d84e804a9ab - WebServer-SG

Outbound rules (1/1)

sg-0127205851ad2229f

Clear filters

< 1 >

⚙

☒	Name	Security group rule ID	IP version	Type	Protocol
☒	-	sg-0127205851ad2229f	IPv4	All traffic	All

STEP 9: Create ec2 security group

EC2 > Security Groups

AWS Global View

Events

▼ Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances

▼ Images

- AMIs
- AMI Catalog

▼ Elastic Block Store

- Volumes

Security Groups (2)

sg-0cbc75d84e804a9ab

Clear filters

< 1 >

⚙

☐	Name	Security group ID	Security group name	VPC ID
☐	-	sg-0cbc75d84e804a9ab	WebServer-SG	vpc-0042a04f0c653ff45
☐	-	sg-03dafd13cde760c94	Database-SG	vpc-0042a04f0c653ff45

Inbound Rule:

- Spot requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager
- ▼ Images
 - AMIs
 - AMI Catalog
- ▼ Elastic Block Store

Inbound rules

Inbound rules (1/1)

sg-09feefa3367a1b49a

Clear filters

< 1 >

⚙

☒	Name	Security group rule ID	IP version	Type	Protocol	Port
☒	-	sg-09feefa3367a1b49a	IPv4	HTTP	TCP	80

Outbound Rule:

- Dedicated Hosts
- Capacity Reservations
- Capacity Manager
- ▼ Images
 - AMIs
 - AMI Catalog
- ▼ Elastic Block Store
 - Volumes

Outbound rules

Outbound rules (1)

sg-0127205851ad2229f

Clear filters

< 1 >

⚙

☐	Name	Security group rule ID	IP version	Type	Protocol	Port
☐	-	sg-0127205851ad2229f	IPv4	All traffic	All	All

STEP 10: Create Relational security group

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Security Groups (1)

Find security groups by attribute or tag

sg-03dafd13cde760c94

Clear filters

1

	Name	Security group ID	Security group name	VPC ID
	-	sg-03dafd13cde760c94	Database-SG	vpc-0042a04f0c653ff45

Inbound Rule:

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

sg-03dafd13cde760c94 - Database-SG

Inbound rules (1)

Search

1

	Name	Security group rule ID	IP version	Type	Protocol
	-	sgr-0116f21ac8f263d6c	-	MYSQL/Aurora	TCP

Outbound Rule:

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

AMIs

AMI Catalog

Elastic Block Store

sg-03dafd13cde760c94 - Database-SG

Outbound rules (1)

Search

1

	Name	Security group rule ID	IP version	Type	Protocol
	-	sgr-0d56c709470bbc063	IPv4	All traffic	All

STEP 11: Create Database

Aurora and RDS

Databases

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Databases (1)

Filter by databases

1

	DB identifier	Status	Role	Engine	Upgrade rollout order	Region
	mydb-instance	Available	Instance	MySQL Co...	SECOND	ap-south-1b

STEP 12: Create Subnet Group

Aurora and RDS

Databases

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

Events

Event subscriptions

Aurora and RDS

Subnet groups

my-db-subnet-2

my-db-subnet-2

Subnet group details

VPC ID

vpc-0042a04f0c653ff45

ARN

arn:aws:rds:ap-south-1:074139044318:subgrp:my-db-subnet-2

Supported network types

IPv4

Description

this is project

Subnets (2)

Availability zone	Subnet name	Subnet ID	CIDR block
ap-south-1a	my-Public-Subnet-1	subnet-0351cc8533bef08dc	10.0.1.0/24
ap-south-1b	my-Private-Subnet-1	subnet-08a8d36f960b75043	10.0.2.0/24

STEP 13: Final Output

Welcome to My Scalable Web App



This project demonstrates EC2, Auto Scaling, ELB, RDS, and S3 integration.

Click Me!

STEP 14: Before Load
Instance:

EC2

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Fast Restore

Instances (1/1)

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Saved filter sets

Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance ID = i-062c49daca4ed7290

Clear filters

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public
☒	WebServer-1	i-062c49daca4ed7290	Running	t3.micro	3/3 checks passed	View alarms	ap-south-1b	-

Auto Scaling Group:

EC2 > Auto Scaling groups

Capacity Manager

Images

Elastic Block Store

Auto Scaling groups (1/1)

Last updated less than a minute ago

Launch configurations

Launch templates

Actions

Create Auto Scaling group

Search your Auto Scaling groups

☒	Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Ma
☒	my-asg	web-template Version 3	3	-	2	1	3

Target Group:

EC2 > Target groups

Elastic Block Store

Network & Security

Load Balancing

Auto Scaling

Target groups (1/1)

Info | What's new?

Filter target groups

☒	Name	ARN	Port	Protocol
☒	my-target-group	arn:aws:elasticloadbalancin...	80	HTTP

Target group: my-target-group

☐	Instance ID	Name	Port	Zone
☐	i-007389cdf5c693021	-	80	ap-south-1a (a...
☐	i-0e8fee15ddbcb610	-	80	ap-south-1b (a...
☐	i-0e08d9d5d1ef58685	-	80	ap-south-1a (a...

